Freya A. Sutedja

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Education

University of California, Berkeley Graduating May 2028
B.A. Applied Mathematics (Data Science) GPA: 4.00
Relevant Coursework: Data Science, Probability & Statistics, Computational Structures, Data Structures & Algorithms

Experience

Mathematics Tutor Sept 2025 – June 2026
PCC Math Success Center Pasadena, U.S.A.

- Delivered **200+ hours** of tutoring in Calculus I–III, Discrete Mathematics, Linear Algebra, and Differential Equations
- Led embedded sessions twice a week in Calculus I, supporting ∼**35** students alongside the instructor

PT Indodax Nasional Indonesia July 2023 – Aug 2023
IT Security Intern Jakarta, Indonesia

- Mapped Nessus scan findings to ∼**270** servers and public IPs to prioritize internet-facing Critical/High risks
- Sorted ∼**100** Cloudflare attacks (SQLi, XSS, brute force) by origin and IP reputation to inform IP blocking

Projects

Steam Game Success Predictor - In Progress

- Cleaned **3 Steam datasets (125K+ games)**, merged 2 on App_ID into a **1.4K-game** table with one-hot encoded tags
- Engineered a composite success score (popularity, reception, engagement) using log transforms, **Bayesian-smoothed** review ratios, and min-max scaling to correct for skew and low-review bias
- Building an NLP pipeline with sentence embeddings (MiniLM) that maps user-entered game ideas to Steam tags and descriptions to predict success

Basketball Free-Throw Form Analyzer

- Extracted 33 body keypoints per frame with MediaPipe and OpenCV, with temporal smoothing and confidence filtering; validated on **5 clips** with **99.3%** of frames detecting the player
- Built the **10-clip** evaluation set and hand-labeled release-frame answer key used to measure form-analysis app's accuracy, plus an OpenCV frame-viewer tool

Overcount: Combinatorics & Probability Practice Site

- Designed and shipped a live practice site (React, TypeScript, Python, Gemini API) end-to-end
- Built a pipeline that verifies Gemini-drafted problems by brute force, with a Monte Carlo check, rejecting **4 of 15** drafts
- Published **528** verified problem instances across an **8-stage** roadmap and **19** counting methods, avoiding per-user AI costs

Leadership & Activities

Technovation Girls
Student Ambassador, International Quarterfinalist

- Co-developed a student community app in FlutterFlow with clubs, note-sharing, a forum, and counseling services
- Led UI/UX and branding, conducted user research, and wrote a business plan with 5-year financial projections

CodeSquad
Co-founder & Volunteer

- Designed the curriculum and built all lesson slides for a weekly Python class serving **10+** underprivileged youth
- Led interactive sessions, mentored students 1:1, and coordinated logistics with partner organizations

Technical Skills

Languages: Python, C++, Java, SQL, MATLAB, HTML/CSS
Frameworks & Libraries: NumPy, Pandas, Matplotlib, Seaborn, OpenCV, YOLOv5, MediaPipe, JavaFX
Tools & Platforms: Git/GitHub, Jupyter, VS Code, LaTeX, Figma, Microsoft Office, Vercel, Gemini API
Spoken Languages: English (Native/Bilingual), Indonesian (Native/Bilingual)